# Automotive Cyber Security Academy 2023

## Basic Information Technology Security

## Topic 6: Hack Yourself

Automotive Cyber Security Academy

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dei departamento de engenharia informática

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# Outline

- Goals
- Software needed
- Identification
- Sniffing
- Live Demo

# Goals

- Discover the open ports that are running in a device (mobile phone, tablet or other)
- Sniff some of the communications that are occurring in your system.

# Software

- **Nmap**, available at: https://nmap.org/download.html
  - Choose the one correct for your platform
  - Used to identify the open ports
- **Wireshark**, available at: https://www.wireshark.org/
  - Choose the one correct for your platform
  - Used for sniffing
- **Netcat**, available at: https://nmap.org/ncat/
  - It might be already installed on your system (“nc” or “netcat”)
  - Useful to send messages, to test protocols

# Identification

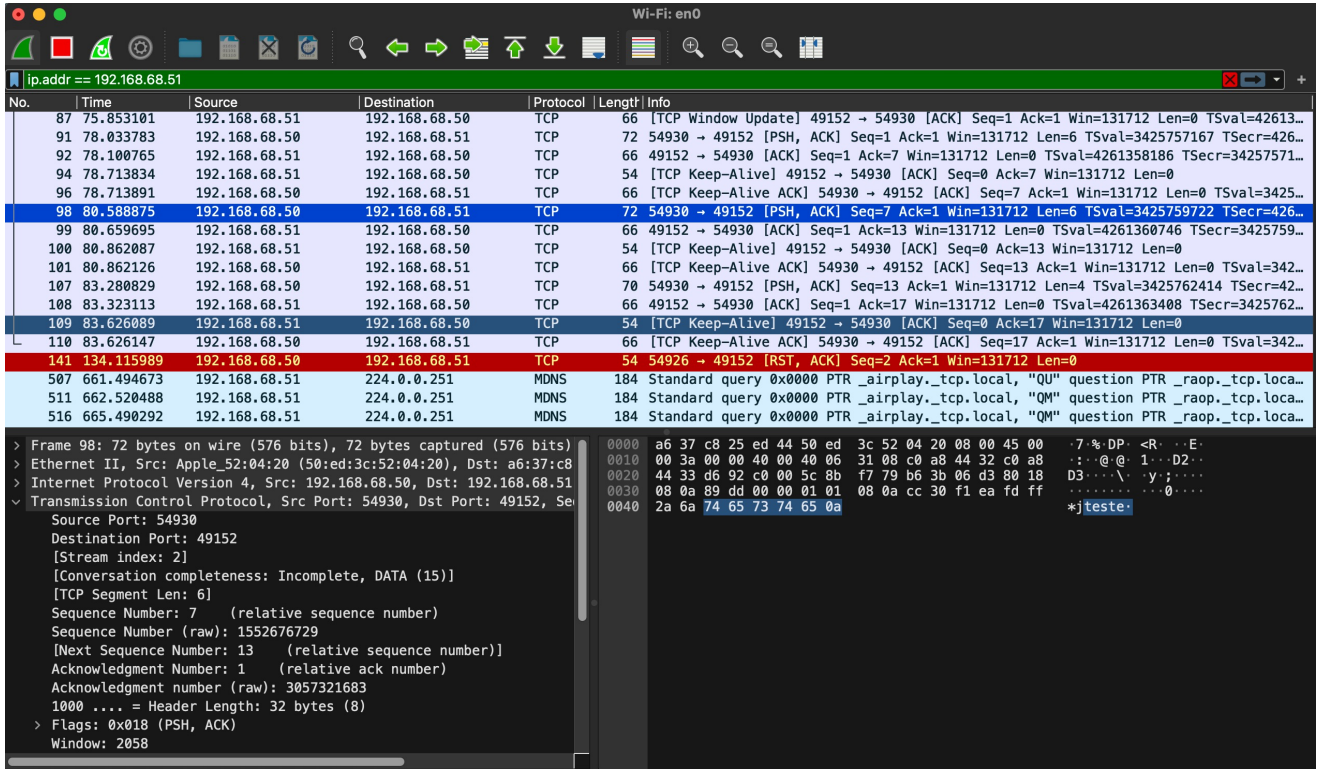
- Use **Nmap**
- Goal:
  - Identify the open ports
- Use the Intense scan, all ports
- Analyse the output

```
Starting Nmap 7.93 ( https://nmap.org ) at 2023-04-29 20:00 WEST
NSOCK ERROR [4.8890s] ssl init helper(): OpenSSL legacy provider failed to load.

NSE: Loaded 155 scripts for scanning.
NSE: Script Pre-scanning.
Initiating NSE at 20:00
Completed NSE at 20:00, 0.00s elapsed
Initiating NSE at 20:00
Completed NSE at 20:00, 0.00s elapsed
Initiating NSE at 20:00
Completed NSE at 20:00, 0.00s elapsed
Initiating ARP Ping Scan at 20:00
Scanning 192.168.68.51 [1 port]
Completed ARP Ping Scan at 20:00, 0.16s elapsed (1 total hosts)
Initiating Parallel DNS resolution of 1 host. at 20:00
Completed Parallel DNS resolution of 1 host. at 20:00, 0.01s elapsed
Initiating SYN Stealth Scan at 20:00
Scanning 192.168.68.51 [65535 ports]
Discovered open port 49152/tcp on 192.168.68.51
Discovered open port 62078/tcp on 192.168.68.51
Completed SYN Stealth Scan at 20:01, 29.07s elapsed (65535 total ports)
Initiating Service scan at 20:01
Scanning 2 services on 192.168.68.51
Completed Service scan at 20:03, 158.74s elapsed (2 services on 1 host)
Initiating OS detection (try #1) against 192.168.68.51
Retrying OS detection (try #2) against 192.168.68.51
Retrying OS detection (try #3) against 192.168.68.51
Retrying OS detection (try #4) against 192.168.68.51
Retrying OS detection (try #5) against 192.168.68.51
NSE: Script scanning 192.168.68.51.
Initiating NSE at 20:04
Completed NSE at 20:04, 14.30s elapsed
Initiating NSE at 20:04
Completed NSE at 20:04, 1.06s elapsed
Initiating NSE at 20:04
Completed NSE at 20:04, 0.01s elapsed
Nmap scan report for 192.168.68.51
Host is up (0.0081s latency).
Not shown: 65533 closed tcp ports (reset)
PORT      STATE SERVICE    VERSION
49152/tcp open  unknown
62078/tcp open  tcpwrapped
MAC Address: A6:37:C8:25:ED:44 (Unknown)
No exact OS matches for host (If you know what OS is running on it, see https://nmap.org/submit/ ).
TCP/IP fingerprint:
OS:SCAN(V=7.93%E=4%D=4/29%OT=49152%CT=1%CU=39803%PV=Y%DS=1%DC=D%G=Y%M=A637C
OS:8%TM=644D6A31%P=x86 64-apple-darwin17.7.0)SEQ(SP=105%GCD=1%ISR=10C%TI=Z%
OS:CI=RD%TS=21)SEQ(SP=103%GCD=1%ISR=107%TI=Z%CI=RD%II=RI%TS=21)OPS(O1=M5B4N
OS:W6NNT11SLL%O2=M5B4NW6NNT11SLL%O3=M5B4NW6NNT11%O4=M5B4NW6NNT11SLL%O5=M5B4
OS:NW6NNT11SLL%O6=M5B4NNT11SLL)WIN(W1=FFFF%W2=FFFF%W3=FFFF%W4=FFFF%W5=FFFF%
OS:W6=FFFF)ECN(R=Y%DF=Y%T=40%W=FFFF%O=M5B4NW6SLL%CC=N%Q=)T1(R=Y%DF=Y%T=40%S
OS:=O%A=S+%F=AS%RD=0%Q=)T2(R=N)T3(R=N)T4(R=Y%DF=Y%T=40%W=0%S=A%A=Z%F=R%O=%R
OS:D=0%Q=)T5(R=Y%DF=N%T=40%W=0%S=Z%A=S+%F=AR%O=%RD=0%Q=)T6(R=Y%DF=Y%T=40%W=
OS:0%S=A%A=Z%F=R%O=%RD=0%Q=)T7(R=Y%DF=N%T=40%W=0%S=Z%A=S%F=AR%O=%RD=0%Q=)U1
OS:(R=Y%DF=N%T=40%IPL=38%UN=0%RIPL=G%RID=G%RIPCK=G%RUCK=0%RUD=G)IE(R=Y%DFI=
OS:S%T=40%CD=S)
```

# Sniffing

- Use **WireShark** and **netcat**
- Goal:
  - Listen to the communications
- Identify the message that is being exchanged, or the communication patterns.



# Live Demo

# Thanks

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The NEW and IMPROVED - ACSA 2023 is here!